

SECTION 33 42 00
SITE STORM UTILITY DRAINAGE PIPING (EMEA)

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PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 1.4C & 1.5C: Replace ‘**LOCAL AHJ**’ with ‘District Administration of Linz-Land (BH Linz-Land, Kärntnerstraße 16, 4020 Linz)’
- 2.1A: Add project specific product information
- 2.5: Schedule updated with project specific product information

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Storm drainage piping, fittings, and accessories.
 - 1. Work extends from a point outside building to detention basin as indicated, and includes chambers, adapters and fittings required for, and final connections to the building storm drainage system.

1.2 RELATED REQUIREMENTS

- A. Section 01 33 00 – Submittal Procedures.
- B. Section 01 73 00 - Execution
- C. Section 01 74 23 - Cleaning
- D. Section 03 30 00 – Cast-in-Place Concrete.
- E. Section 31 23 23 – Fill, Backfill and Compaction.
- F. Section 31 23 16 – Excavation.
- G. Section 31 23 16.13 – Trenching for Site Utilities.

1.3 REFERENCE STANDARDS

Referenced standards shall be the latest editions adopted as of the project bid date, except that if an earlier edition is cited in a code or law adopted by the local AHJ, that edition shall take precedence.

- A. Comply with the following:
 - 1. Ductile iron/cast iron pipes and fittings for drainage
 - a. All pipes and fittings shall comply with the latest European standards as follows:
 - 1) Ductile iron pipes and fittings to EN 598 or EN 545.
 - 2) Iron spigot and socket drain pipes to EN 877.
 - 2. Concrete pipes and fittings:
 - a. Unreinforced and reinforced concrete pipes and fittings shall comply with the requirements of EN 1916 and EN 681.

- b. The cement used in their manufacture shall be ordinary, rapid-hardening or sulphate resisting (as required) Portland cement. Samples of the aggregate used in the manufacture and certificates of the date of manufacture shall be submitted to the A/E of Record when required.
 - c. Pipe Class shall be "extra strength" pipes throughout.
 - d. Concrete pipes shall be spigot and socket with flexible joints of make and type approved by the A/E of Record.
- 3. Joint rings and lubricants
 - a. Coupling sealing rings shall be manufactured from elastomers, conforming to the performance requirements of EN 681. All seals to be EPDM (ethylene propylene diene monomer) or other as specified by the A/E of Record.
 - b. Joint lubricants for sliding joints shall have no deleterious effect on either the joint rings or pipes and shall be unaffected by the liquid to be conveyed.
- 4. PE twin walled drainage pipes
 - a. PE twin walled drainage pipes shall comply with EN 13476. Alternatively pipes similar or approved may be accepted.
- 5. Vitrified clay drainage pipes
 - a. Vitrified clay drainage pipes shall comply with EN 295. Alternatively pipes similar or approved may be accepted.

1.4 QUALITY ASSURANCE

- A. Services of an Inspection and Testing Agency will be used for this Work.
- B. Use adequate numbers of skilled workmen who are thoroughly trained and experienced in the necessary skills and who are completely familiar with the specified requirements and the methods needed for proper performance of the work of this Section.
- C. District Administration of Linz-Land (BH Linz-Land, Kärntnerstraße 16, 4020 Linz) ~~LOCAL AHJ~~ rules governing storm sewer service and standard details for storm sewers.
- D. All products shall be CE marked to indicate conformity with health, safety, and environmental protection standards for products sold within the European Economic Area.

1.5 SUBMITTALS

- A. Before any equipment is installed, submit and obtain approval from the Authority Having Jurisdiction (AHJ). AHJ approval shall be included with shop drawing submittals.
- B. Shop Drawings and Product Data Catalog Cuts to be submitted at one (1) time. Partial submissions will not be accepted.
- C. Shop Drawings:
 - 1. After approval by District Administration of Linz-Land (BH Linz-Land, Kärntnerstraße 16, 4020 Linz) ~~LOCAL AHJ~~, submit a copy of approved submittals, product catalog cuts, and details to Design Professional for his approval before starting Work. Shop Drawings and product catalog cuts to be submitted at one (1) time. Partial submissions will not be accepted.

- D. Product Data: For each product indicated.
- E. Record documents meeting the following Minimum Data Collection Requirements for Stormwater Record Drawings shall be stamped and submitted for Engineering approval. Refer to Civil As-Built Standard Guide (UID: 16232) for submittal requirements.
 - 1. Minimum requirements for gravity storm sewers:
 - a. Provide survey grade coordinated global positioning system (GPS) or field survey electronic data horizontal and vertical coordinates for each manhole, drainage structure, headwall, outlet structure, and so on, location.
 - b. Provide ground elevation, top elevation and invert elevations for each manhole, drainage structure, cleanout, headwall, outlet structure, trench drain, and so on.
 - c. Provide horizontal and vertical coordinates along linear installations at an interval not less than once per 30 m of alignments. Include coordinates for any downpipe connections or service laterals.
 - d. Stormwater control structures shall require a field sketch with geometry and invert coordinates.
 - 2. Minimum requirements for utility crossings during stormwater utility installation:
 - a. If a new utility installation requires the crossing of an existing utility within the site, the project record drawings shall indicate the location, size, and elevation at which the existing utility was incurred.
 - b. Provide survey grade coordinated global positioning system (GPS) or field survey electronic data horizontal and vertical coordinates for each utility visually identified during the installation of the subject utility.
 - c. Temporary utilities can be omitted from this requirement.
 - 3. Record documents shall include noted plans indicating at a minimum the information listed in Items 1 and 2 above and shall be accompanied with electronic GPS coordinate file, or survey ASCII file in electronic format. Field notes shall be provided upon request. AutoCAD files should be submitted ideally also.
- F. Quality Control Procedures:
 - 1. Testing procedures for new piping for approval.
 - 2. Site test reports: Hydrostatic test reports.

1.6 PROTECTION OF OTHER UTILITIES

- A. Location:
 - 1. Approximate location of certain known underground lines are identified on the project drawings.
 - 2. Existing small lines not shown.
 - 3. Locate small and other possible utility lines using trial pits, electronic pipe finder, or other approved method.
 - 4. Excavate and expose existing underground utilities ahead of trenching operations.
- B. Repair or replace any damaged utility line or structure at no additional cost to Owner.

1.7 CONFLICTING UTILITIES

- A. Where alterations to existing utilities are shown to avoid conflicts, make alterations at no

cost to Owner.

- B. Remove and/or relay conflicting utilities, when so directed by the A/E of Record, at the expense of the Owner.

PART 2 PRODUCTS

2.1 UNDERGROUND PIPE AND FITTINGS

- A. Shall be of sizes and material indicated on Drawings and shall conform to Piping Material Schedule A(SU).

Primary storm water drainage network pipework (≤ 300 mm Diameter):

- Compliant with EN 1852
- Material: Polypropylene (PP)

Primary storm water drainage network pipework (> 300 mm Diameter):

- Compliant with EN 1916 and EN 681
- Material: Reinforced concrete

Joints:

- Push fit / PP watertight spigot and socket joints compliant with EN 1852 and EN 681
- Push fit / Spigot and socket joints compliant with EN 1916 and EN 681

Jointing material:

- Rubber compression type gasket compliant with EN 1852 and EN 681

2.2 MANHOLES, CATCH PITS AND KERB INLETS

- A. Construct manholes of precast reinforced concrete manhole sections meeting EN 1917. Provide suitable rubber gasket joints, meeting EN 681-1, between sections. Pipe to manhole flexible seals shall conform to EN 681-1. Manhole bases may be precast or cast-in-place concrete. Top section of manhole shall be eccentric cone type so that manhole steps form a straight ladder.
- B. Construct manholes of polyethylene (PE) and polypropylene (PP) manhole sections meeting EN 13598. Pipe to manhole flexible seals shall conform to EN 681-1. Manhole bases may be precast or cast-in-place concrete. Top section of manhole shall be eccentric cone type so that manhole steps form a straight ladder. A concrete surround shall be used when in carriageways and where high groundwater is present to prevent flotation. PE manholes should be used as the preferred option wherever feasible.
- C. Refer to manhole typical details as shown on the project drawings for minimum sizes for various pipe diameters. Regardless of the minimum size the manhole shall be sized to provide the minimum clear landings required and set out in the typical details.
- D. Refer to manhole typical details for requirements for step iron/ladder access.

- E. Provide invert channels of cast-in-place high strength concrete topping (hard aggregate wearing screed to EN 13813 and EN 13813) shall be provided, laid and finished in accordance with this specification. Concrete or equivalent.
 - 1. Hard aggregate wearing screed concrete shall be strength Class C32/40 with maximum aggregate size 4/10 mm. It shall conform to EN 206.
 - 2. Aggregate shall be crushed granite complying with EN 12620 for heavy duty concrete floor finishes. The aggregate shall comply with either the grading requirements in Table 6 for 0/10 GA90 or 0/6.3 GA90 maximum size, all-in aggregate, or the coarse aggregate grading in Table 2 for 4/10 GC85/20 single sized aggregate and the sand within the limits of 0/4 (CP) or 0/2 (MP) of EN 12620.
 - 3. Hard aggregate wearing screed concrete to this Clause shall be laid on base concrete which has been prepared in the following manner. The laitance on the base concrete shall be removed by wire brushing and by hosing with water. This shall be done after the initial set of the base concrete but before hardening and without disturbing the aggregate. Hard aggregate wearing screed concrete to this clause shall be laid not later than three days after placing the base concrete.
 - 4. After compaction the surface of the render or screed shall be trowelled to a smooth finish to the required levels, profiles and contours and shall present a surface finish as follows:
 - a. The concrete shall be leveled and screeded to produce a uniform surface to the profile shown on the Drawings. When the moisture has disappeared and the concrete has hardened sufficiently to prevent laitance from being worked to the surface, the finish shall be steel trowelled under firm pressure to produce a dense, smooth uniform surface free from trowel marks.
- F. Mild steel safety chains shall be provided as required and set out in the typical details. They shall be in accordance with EN 818-3 Grade 4 nominal size 8 by 24. After manufacture, mild steel safety chains shall be hot dip galvanized in accordance with EN ISO 1461.
- G. Stainless steel safety chains shall be manufactured from Grade X5CrNiMo 17-12-2 steel conforming to EN 10088-3. Chain links shall be welded and have an internal length not exceeding 45 mm and an internal width of between 12 mm and 18 mm. The fins caused by welding shall be removed and the weld shall be smoothly finished all round.
- H. The chain shall have a breaking force of 30 kN and a proof force of 15 kN when tested in accordance with EN 818-1.
- I. Anchor bolts for fixing safety chains shall be of the stainless steel safety type which provides a progressive mode of failure.
- J. Provide drop connections where the difference between inflow and outflow elevation exceeds 750 mm. Encase drop pipe and fittings in C30/35 concrete suitable for local ground conditions.

2.3 ROAD GULLIES

- A. Precast concrete gullies shall comply with the requirements of EN 1917, trapped with capped rodding eye, 450 mm diameter and 750 mm deep. Gullies shall be set on a 150 mm

bed and surrounded with a minimum of 150 mm Grade C16/20 Dmax20 Concrete.

- B. Vitrified clay gullies shall not be used.
- C. Gully frames shall be set on at least one course of engineering brickwork to finish with the top of the slope at not more than 6 mm below finished road surface. No brick packing shall extend in height for more than three brick courses.
- D. Cast iron gully covers, gratings and frames to be a minimum Class D400 and shall comply with the relevant provisions of EN 124.

2.4 KERB DRAINS

- A. Blocks to comply with EN 1433.
- B. Loading class in accordance to EN 124 with a minimum of D400.

2.5 SITE STORM SEWER (BELOW GROUND) – SCHEDULE A(SU)

Pressure/Temperature Limits: Atmospheric/Ambient

General Materials: Concrete, Polypropylene

Item	Pipe Sizes	Description	Reference Standards
Pipe	375 mm and larger	Reinforced concrete, storm drain and sewer pipe	EN 1916 and EN 681
Pipe	300 mm and smaller	Polypropylene, storm drain and sewer pipe	EN 1852
Joints	375 mm and larger	Push-on or bell and spigot	EN 1916 and EN 681
Joints	300 mm and smaller	Push fit / PP watertight spigot and socket joints	EN 1852 and EN 681
Jointing material	375 mm and larger	Rubber compression type gasket	EN 681
Jointing material	300 mm and smaller	Omni -directional rubber gasket for PE PP pipe	EN 1852 and EN681

2.6 FRAMES, COVERS AND GRATINGS

- A. Manhole frames and covers shall be gray cast iron meeting a minimum EN 124 Group 4 Class D400. Covers shall have words "STORM SEWER" cast in center. Bearing surfaces for frames and covers shall be machine finished to prevent covers from rocking and jamming in any position. Grind smooth all projections and roughness. No bituminous coating shall be applied.

2.7 RIP RAP STONE

- A. Rip rap stone for anti-scour on outlet structures and where velocities are greater than 1 m/s shall be durable rock sized as indicated on drawings and reasonably well graded.

2.8 BURIED UTILITY WARNING AND IDENTIFICATION TAPE

- A. Provide polyethylene plastic tape manufactured specifically for warning and identification of buried utility lines.
- B. Tape shall be of type provided in rolls, 150 mm minimum width, color-coded for utility involved, with warning and identification imprinted in bold black letters continuously and repeatedly over entire tape length.
- C. Warning and identification of buried storm sewer lines shall read "CAUTION - STORM SEWER LINES BURIED BELOW" or similar.
- D. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

2.9 TRENCH/CHANNEL DRAINS

- A. Trench frame and grate shall be a minimum of EN 124 Group 4 Class DN40.
- B. Trench drains shall slope to outlet at a minimum of 0.5 percent.
- C. Provide coatings as required for fuel/ chemicals.
- D. Where located in cold climates trench/channel drains shall have trace heating to prevent freezing.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Inspection: Give two (2) weeks written advance notice to Inspection and Testing Agency, AHJ if required, Owner's Representative and Design Professional before laying pipe, so that inspection can be scheduled.
- B. General: Install Work in accordance with Manufacturer's written instructions, approved submittals, referenced sections of Specifications and Drawings, and Codes and Standards.
- C. Watertight construction: Take all precautions necessary to obtain watertight construction of all Work.
 - 1. Apply joint sealer or ring gasket to wall sections, set firmly in place to assure watertight joints.
 - 2. Non-shrink grout to be applied where any concrete pipe connects to a concrete surface of a structure, and any seams between structures where two or more structures are joined together. Application of product shall cover the full perimeter of the connection and seam, on both the interior and exterior of the pipe and structure.
- D. Lines and grades: Stake out Work at least one (1) week prior to installation to allow for inspection. Set all stakes for lines and grades. Offset stakes shall normally be set at intervals of 15 meters and at manhole locations. On steep slopes, intervals shall be decreased to 7.5 meters or as directed by Design Professional.
- E. Tolerance: Do not allow horizontal and vertical alignment of trenched gravity sewer lines to vary from design line and grade at any point along the pipe by more than 1 percent of the inside diameter of the pipe or 5 mm, whichever is larger. Reverse slope is prohibited. Remove and reinstall pipe to proper grade when tolerances are exceeded.
- F. Trench excavation: Refer to Section 31 23 16.13 - Trenching for Site Utilities.
- G. Existing lines and structures: Verify the inverts and locations of existing and proposed piping, manholes and other structures before installing any materials. In case of discrepancy from what is indicated on Drawings, request further written instruction from A/E of Record before installation starts.
- H. Clearances: Maintain a minimum of 150 mm vertical separation at water main crossings. Where the storm sewer crosses over or less than 450 mm below a water main, locate one full length of sewer pipe of water main material so both joints are as far as possible from the water main. The storm sewer and water pipes must be adequately supported and have watertight joints. Use a low permeability soil for backfill material within 3 m of the point of crossing.
- I. Bedding: Accurately grade pipe trench bed with a minimum of 150 mm of utility trench bedding material as specified in Section 31 23 23 - Fill, Backfill and Compaction.
- J. Manholes, catch pits and kerb inlets: Install on minimum 150 mm of compacted granular

material as specified in Section 31 23 23 - Fill, Backfill and Compaction.

1. Set base level so that walls will be plumb.
2. Apply joint sealer to wall sections, set firmly in place to assure watertight joints.
 - a. All structures, and their respective pipe connections, shall be constructed so that water tightness is obtained.
3. Non-shrink grout to be applied where any concrete pipe connects to a concrete surface of a structure, and any seams between structures where two or more structures are joined together. Application of product shall cover the full perimeter of the connection and seam, on both the interior and exterior of the pipe and structure.
4. Form manhole invert channels directly in the concrete of the manhole base, with mortar, or by laying full section sewer pipe through the manhole and breaking out the top half after surrounding concrete has hardened. Smooth the floor of the manhole outside the channels, and slope toward the channels at not less than 1:30 nor more than 1:10.
 - a. Shape the invert channels to be smooth and semicircular, conforming to the inside of the adjacent sewer section.
 - b. Make changes in direction of flow with a smooth curve of as large a radius as the size of the manhole will permit.
 - c. Make changes in size and grade of channels smoothly and evenly.
 - d. Slope invert uniformly from invert of inlet to invert of outlet.

K. Pipe and fittings:

1. Install and join pipe in accordance with Manufacturer's written instructions. Pipe shall be laid true to line and made with uniform bearing under full length of barrel of pipe. Pipe laying shall proceed upgrade.
2. Cap or plug open ends of pipe to prevent entry of animals or foreign material. Do not permit groundwater or rainwater to enter system. Inside of pipe and fittings shall be free of foreign material, and outside of spigot (or tongue and spigot) and inside of bell (or tongue and groove) shall be clean.
3. Carefully lower pipe and accessories into trench using suitable tools and equipment. Do not drop or dump materials into trench.
4. Perform pipe cutting by machine in such manner so as not to cause damage to pipe and to leave a smooth end at right angles to axis of pipe.
5. Lay each pipe accurately to the indicated line and grade, aligning so the sewer has a uniform invert.
6. Continually clear interior of the pipe free from foreign material.
7. Before making pipe joints, clean and dry all surfaces of the pipe to be joined.
8. Use joint primers as recommended by the Pipe Manufacturer.
9. Place, fit, join and adjust the joints to obtain the degree of water tightness required.

L. Backfilling and compaction: Refer to Section 31 23 23 - Fill, Backfill and Compaction.

M. Warning tape: Lay warning tape in backfill 200 mm to 300 mm below grade.

N. Penetrations: Where pipe penetrates concrete walls or slabs, provide watertight pipe sleeve-seal assembly around piping.

3.2 TESTING

A. Testing of drains and connections

1. Testing of drains and connections is to be carried out in accordance with EN 1610 and EN 752 Parts 1-4.
2. Before testing any pipeline, the Contractor shall ensure that it is anchored adequately and that thrusts from bends, branch outlets or from the pipeline ends are transmitted to solid ground or to a suitable temporary anchorage.
3. Adequate safety signs indicating "test in progress" shall be provided for the duration tests.
4. Open ends shall be stopped with plugs, caps or blank flanges properly jointed.
5. All jointed pipe drains shall be tested after being laid and jointed and before surrounding and backfilling is commenced to ensure that the jointing is satisfactory.
6. No pipes are to be covered up until inspected by the A/E of Record.
7. The test shall be repeated after the completion of backfilling and construction. The A/E of Record will also require a further test to be carried out on completion of the Contract prior to the commencement of the Maintenance Period.
8. Drains/sewers up to and including 750 mm diameter shall be tested by means of an air or water test as instructed by the Employers Representative and drains/sewers greater than 750 mm diameter by visual inspection.

B. Air testing for gravity sewer/drains

1. When an air test is used, the apparatus shall be of a type approved by the A/E of Record and the test shall in all cases be applied in the presence, and to the satisfaction of the A/E of Record or his representative.
2. Air shall be pumped into the length of pipes under test until a pressure equivalent to 100 mm of water is indicated and maintained without loss exceeding 25 mm for five minutes without further pumping.
3. Concrete pipes may be damped prior to testing. Failure to pass the test shall not preclude acceptance of the pipeline if a successful water test can subsequently be carried out.
4. In the event of the failure of such tests, or any infiltration occurring, the Contractor will be required to locate the defect and make good by providing and laying new pipes or otherwise repairing the leak as directed by the Engineer of Record.
5. All instruments used to perform the tests shall be calibrated, and a calibration certificate shall be made available for inspection.

C. Water test for gravity sewers/drains

1. The test pressure for gravity sewers shall not be less than 1.2 m head of water above the pipe soffit or ground water level, whichever is the higher, at the highest point and not greater than 6 m head at the lowest point of the section.
2. Steeply graded sewers shall be tested in stages in cases where the maximum head, as stated above, would be exceeded if the whole section were tested in one length.
3. The pipeline shall be filled with water and a minimum period of 2 hours shall be allowed for absorption, after which water shall be added from a measuring vessel at intervals of 5 minutes and the quantity required to maintain the original water level noted. Unless otherwise specified, the length of pipeline shall be accepted if the quantity of water

added over a 30 minute period is less than 0.5 liters per linear meter per meter of nominal diameter.

4. Notwithstanding the satisfactory completion of the above test if there is any discernible leakage of water from any pipe or joint, the pipe shall be replaced and/or the joint remade, as appropriate by the Contractor, and the test repeated until leakage is stopped.

D. Infiltration

1. Gravity sewers/drains and manholes shall be inspected and tested as appropriate for infiltration, after backfilling. All inlets to the system shall be effectively closed, and any residual flow shall be deemed to be infiltration.
2. The sewers/drains and manholes shall be accepted as satisfactory if the infiltration, including infiltration into manholes, in 30 minutes does not exceed 0.5 liter per linear meter per meter of nominal diameter.
3. Notwithstanding the satisfactory completion of the above inspection or test, if there is any discernible flow of water entering the sewers/drains or manholes at a point which is located either by visual or CCTV inspection, the Contractor shall take such measures as are necessary to stop such infiltration.

3.3 INSPECTION

A. General:

1. Clean and prepare for observation of each block or section of sewer upon completion, or at such other time as the A/E of Record may direct.
2. Each section between manholes shall show a full circle of light when viewed from either end.
3. Repair all visible leaks.
4. Correct broken or cracked pipe, mislaid pipe and other defects.
5. All repairs, relaying of sewers, and so on, required to bring the sewers to specified status shall be made at no additional cost to the Owner.

B. Provide two (2) weeks written advance notice to the Owner's Representative and Design Professional before inspection.

C. Video inspect all storm sewer network. Work should be carried out according to local specifications for CCTV condition surveys.

D. Provide digital copies of the inspection video and report to the Owner's Representative and Design Professional.

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	August 2024	Initial version – BOD Draft submission
0.2	September 2024	BOD Final Submission
0.3	February 2025	Draft IFC Updates
1.0	12 th of March 2025	Final IFC Issue